Equivalence-principle classification of dark-sector mechanisms in an emergent-gravity substrate: a vestigial-only violation surface

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Dark-sector mechanisms in candidate emergent-gravity scenarios have historically been discussed in prose without explicit commitment to their equivalence-principle (EP) status. This paper makes that commitment formal: we classify six distinct mechanisms relevant to a recent emergent-gravity Anderson-Higgs-type Wilsonian (ADW) program by which level of the equivalence principle they violate, encode the classification as decidable predicates in Lean 4, and prove the structural consequences. Two mechanisms violate the weak EP and hence all three levels: *vestigial differential coupling*, in which the metric and tetrad fields acquire different vacuum expectation values so that bosons and fermions couple to gravity through different effective geometries (giving the maximal $\eta = 1$, already excluded by current satellite tests [2]); and *vestigial relics*, which predict a residual $\eta \sim 10^{-18}$ from defect remnants in a full-tetrad phase, sub-MICROSCOPE and at the projected design sensitivity discussed for STEP-class satellite missions. The remaining four mechanisms — Einstein–Cartan torsion-trace coupling, fracton subdiffusion [3], Berezhiani–Khoury Thomas–Fermi superfluid dark matter [4], and hidden-sector $\mathbb{Z}_{16}$ singlets — satisfy all three EP levels. The structural conclusion is that EP violation in this substrate is *vestigial-only*: any positive EP-violation result implies vestigial physics, and any null result at sub-10^{-19} precision excludes vestigial gravity. The classification ships in Lean 4 with 24 substantive theorems, zero `sorry` statements, and zero new axioms beyond the kernel's three.

I. INTRODUCTION

The equivalence principle is the foundational symmetry of classical general relativity [1]. It admits three nested formulations:

- **Weak EP (wep)**: test masses follow the same trajectory in a gravitational field, regardless of internal composition. The violation parameter is $\eta = (a_1 - a_2)/\frac{1}{2}(a_1 + a_2)$. The MICROSCOPE satellite mission [2] sets $|\eta| < 1.0 \times 10^{-15}$, the strongest current bound.
- **Einstein EP (eep)**: WEP together with local Lorentz invariance and local position invariance.
- **Strong EP (sep)**: EEP extended to gravitational self-energy (Nordtvedt parameter η_N), tested by lunar laser ranging.

A mechanism that violates WEP thereby violates EEP and SEP. The converse is not generally true: a mechanism may violate EEP via Lorentz-invariance breaking in subsystems with degenerate gravitational coupling, or SEP alone via composition-independent self-energy effects. Each mechanism therefore has a *weakest* EP level it violates (or no violation if all three are satisfied), which propagates upward.

The dark-sector landscape of recent emergent-gravity programs contains several distinct mechanisms whose EP-violation status has been discussed only at the prose level. This paper takes six concrete mechanisms — spanning vestigial gravity, Einstein–Cartan torsion, fracton physics, single-field superfluid dark matter, and hidden-sector $\mathbb{Z}_{16}$ singlets — and forces an explicit commitment per mechanism through a Lean 4 formalization. The 6×3 classification matrix is the substantive output: it surfaces both the vestigial-only character of the project's EP-violation surface and the η-scale separations relevant to near-term satellite EP tests.

II. MECHANISM ENUMERATION

The six mechanisms are:

Vestigial differential coupling. In a phase where the metric correlator is non-zero but the tetrad VEV vanishes, bosons couple minimally to the metric $g_{\mu\nu}$, while fermions require a non-zero tetrad VEV $\langle e^a_\mu \rangle$ to couple to gravity. The differential coupling parameter $\Delta_{\mathrm{EP}} = 1 - \langle e^a_\mu \rangle / \sqrt{\langle e^a_\mu e^b_\nu \rangle \eta_{ab}}$ takes the maximal value $\Delta_{\mathrm{EP}} = 1$ in this phase. The corresponding $\eta_{\max} = 1$ is already incompatible with any current EP precision.

Vestigial relics. In the full-tetrad phase, residual differential coupling from topological-defect remnants gives $\eta \sim 10^{-18}$. This is sub-MICROSCOPE and at the projected design sensitivity discussed for STEP-class satellite missions ($\eta \sim 10^{-18}$ target).

Einstein–Cartan torsion-trace coupling. A torsion-loop dark-matter mechanism in Einstein–Cartan gravity. Its failure mode as a cold-dark-matter candidate is kinematic — the equation of state is $w = 1/3$ rather than $w = 0$ — not an EP violation. The torsion-trace loop coupling is universal across charged species, so no species-dependent acceleration arises at tree level.

Fracton subdiffusion. Pretko-style dipole-conservation enforces universal mobility decay across all matter species [3]. The subdiffusion rate is identical for all species (the underlying constraint is a charge-conservation relation), so no WEP violation arises.

Thomas–Fermi superfluid dark matter. Berezhiani–Khoury single-field condensate dark matter [4] with uniform coupling to gravity. The Thomas–Fermi profile is composition-independent; phenomenology appears in galactic merger sonic-boom signatures, not in EP tests.

Hidden-sector $\mathbb{Z}_{16}$ singlets. Standard-Model-singlet dark-matter candidates carrying the $\mathbb{Z}_{16}$ topological charge but no Standard-Model gauge charge. Decoupled from visible-sector EP tests by construction.

III. FORMAL CLASSIFICATION

The classification ships as a Lean 4 formalization in `lean/SKEFTHawking/EquivalencePrinciple.lean`. Three EP levels `WEP`, `EEP`, `SEP` are encoded as an inductive type with a numerical-order projection: `WEP` $\mapsto 0$, `EEP` $\mapsto 1$, `SEP` $\mapsto 2$. Six mechanisms form a second inductive enumeration. The function `violationLevel` : `EPMechanism` $\to$ `Option EPLevel` assigns each mechanism its weakest violated level, or `none`. Decidable predicates `violatesAt` and `satisfiesAt` implement the level propagation: a WEP-violator violates WEP, EEP, and SEP via the order projection.

The substantive theorems are:

- Six per-mechanism classifications (`*_violates_WEP` for the two vestigial mechanisms, `*_satisfies_all_EP` for the remaining four).
- A central structural statement `ep_violation_is_vestigial_only`: among the six mechanisms, exactly the two vestigial-phase phenomena violate WEP.
- A consistency theorem `vestigial_microscope_violation_consistent` pinning the classification's commitment for vestigial gravity to the project's companion module on dark-sector classification.
- Distinguishability theorems `vestigial_vs_fangGu_distinct_EP_profile` and `distinct_violationLevel_implies_distinct_mechanism`.
- A structural lemma `violatesAt_mono` encoding the hierarchical subsumption of EP levels.
- Quantitative anchors tying the classification to the published η-bound constants: `vestigial_phase_eta_violates_microscope_bound` ($\eta_{\max} = 1.0 > 10^{-15}$, `norm_num`-backed) and `vestigial_relics_below_microscope_bound` ($\eta_{\rm relic} = 10^{-18} < 10^{-15}$, also `norm_num`-backed).
- A cross-module bridge `fangGu_failure_mode_is_kinematic_not_ep`, which imports the Einstein–Cartan torsion module and invokes the torsion-trace obstruction theorem to confirm that the failure mode of that mechanism is kinematic rather than EP-related.

A Python mirror at `src/equivalence_principle/` provides the identical classification as a dictionary lookup, 38 `pytest` cases pinning Python and Lean to numerical agreement, and four named numerical constants: `MICROSCOPE_BOUND` $= 1.0 \times 10^{-15}$, `STEP_TARGET` $= 1.0 \times 10^{-18}$, `VESTIGIAL_PHASE_ETA_MAX` $= 1.0$, `VESTIGIAL_RELICS_ETA` $= 1.0 \times 10^{-18}$.

IV. CROSS-MECHANISM STRUCTURE

The matrix surfaces three structural facts:

Vestigial-only violation surface. Among six mechanisms spanning four distinct dark-sector physics families, only the two vestigial-phase phenomena violate the equivalence principle. This is the load-bearing substantive content: any positive EP-violation result at $\eta \in [10^{-19}, 10^{-15}]$ implies vestigial physics within this dark-sector landscape.

Two distinct vestigial η scales. Within the vestigial family, the in-phase mechanism ($\eta_{\max} = 1$, already excluded) and the relic mechanism ($\eta \sim 10^{-18}$, sub-MICROSCOPE) sit at very different scales. The MICROSCOPE bound separates them: vestigial-phase coupling is ruled out at current precision, while vestigial relics await a next-generation satellite test at $\eta \sim 10^{-18}$ precision.

Cross-mechanism distinguishability. The four non-violating mechanisms are degenerate by EP status alone (all satisfy all three levels), so EP tests alone cannot distinguish them; distinguishability requires additional kinematic equation-of-state signatures, cosmological core-profile constraints, or direct-detection floors. EP tests *do*, however, distinguish the vestigial family from the rest.

V. OUTLOOK

A positive next-generation satellite-EP-test result at $\eta \sim 10^{-18}$ would corroborate the vestigial-relics mechanism and place a bound on the underlying defect-remnant density. A null result at $\eta \lesssim 10^{-19}$ excludes vestigial-relic-induced violations and forces a re-examination of which late-time emergent-gravity phase the substrate occupies. The four non-vestigial mechanisms make no testable EP prediction at any precision, so EP tests alone are agnostic to them.

The next steps integrate this classification with cosmological perturbation constraints (different EP profiles produce distinct CMB-polarization signatures via differential coupling of photons versus neutrinos) and with energy-condition analyses, since mechanisms violating SEP may also violate the dominant energy condition on

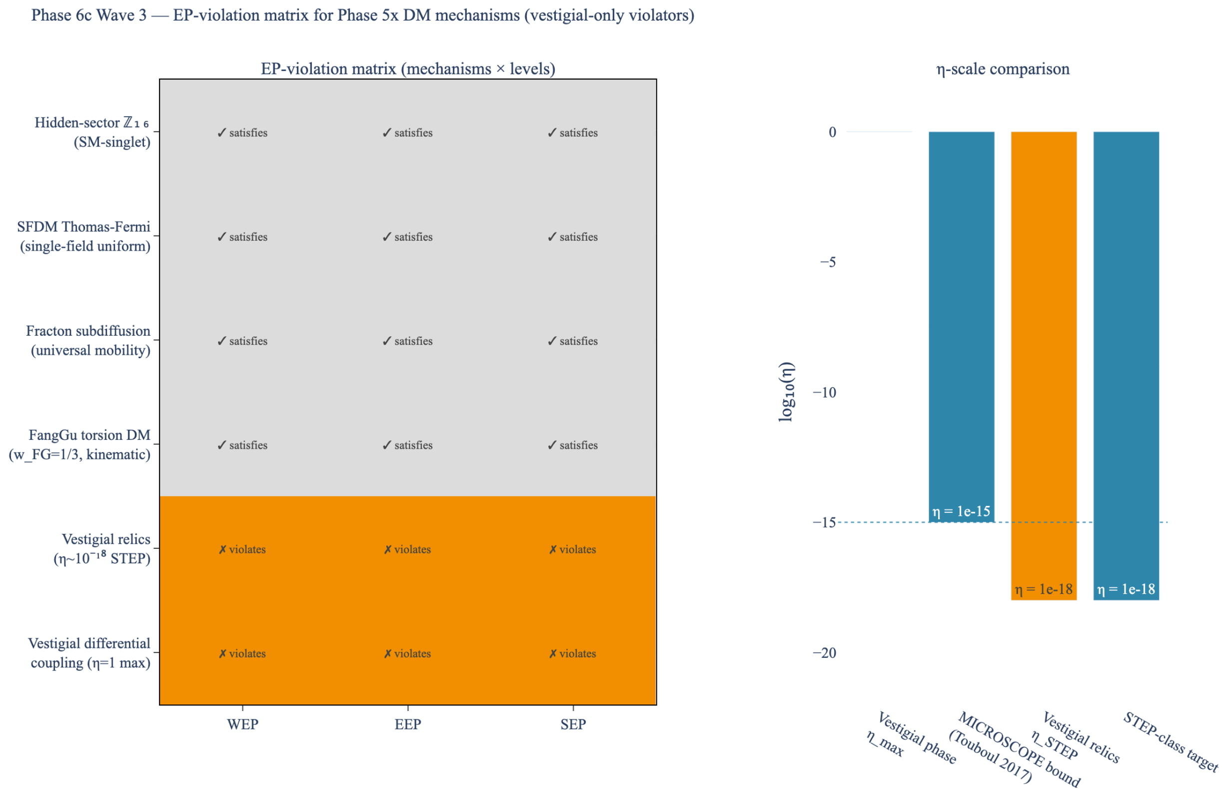

FIG. 1. Equivalence-principle violation matrix (left) and η-scale comparison (right). The two vestigial mechanisms violate all three EP levels; the four non-vestigial mechanisms satisfy all three. The right panel locates the vestigial-phase $\eta_{\max} = 1$, the MICROSCOPE bound $\eta < 10^{-15}$ [2], the vestigial-relic $\eta \sim 10^{-18}$, and the next-generation satellite-test target $\eta \sim 10^{-18}$ on a common log scale.

the relevant emergent stress-energy tensor, opening loopholes for cosmological-singularity theorems.

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